Profile Monitoring

Statistical Quality Control Seminar

Asini Nethma Wanniarachchi
147972

Content

01 Introduction

02 What is Profile Monitoring?

03 Linear Profile Monitoring

04 Phases of Profile Monitoring : Phase I

05 Phases of Profile Monitoring : Phase II

06 Control Charts : Hotelling's T^2 Chart

07 Control Charts : Alternative Charting Strategies

08 Performance Measures

09 Practical Application: R Simulation Setup

10 R Simulation Results: Phase I

11 R Simulation Results: Phase II

12 Conclusion

Introduction

- Statistical Quality Control (SQC) has long been a foundation for manufacturing and service industries, providing tools to monitor, control, and improve process performance.
- Statistical Process Control is a specific branch within SQC, useful in achieving process stability and improving capability through variability reduction.

Problem: Traditional methods are limited for complex processes where quality is defined by a functional relationship between a dependent and one or more independent variables, necessitating more advanced methods.

> *"quality is inversely proportional to variability"*

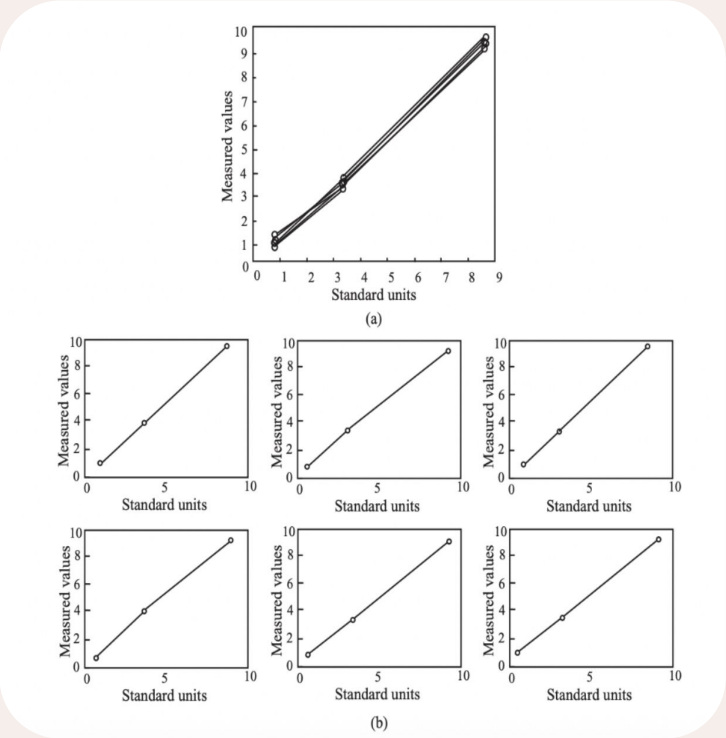
What is Profile Monitoring?

- It is a statistical process control method used to assess the stability of the relationship between a dependent variable and one or more independent variables over time.

- It's necessity arises when changes in the underlying functional relationship, not just individual variable shifts, indicate process degradation or assignable causes.

A "profile" refers to a functional relationship,

typically denoted as $y = f(x;\beta)$



This figure illustrates multiple calibration profiles, both overlaid (a) and in separate plots (b), showing the relationship between measured and standard units.

Linear Profile Monitoring

- They characterize a quality relationship where a single dependent variable (y) follows a straight-line relationship with one or more independent variables (x).
- For example, in a manufacturing process, the hardness of a material (y) might linearly increase with the curing temperature (x).

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

y_i the i-th observed dependent value
x_i is the i-th value of the independent variable
β_0 is the intercept
β_1 is the slope
ϵ_i is the random error term; assumed to be normally distributed

Non linear profiles: Has have a single dependent variable that is related to one or more independent variables which results in a curve

Multivariate Profiles: Has multiple dependent variables and each of these variables is related to one or more independent variables

Binary response profiles: The probability of a specific outcome as a function of one or more independent variables

Geometric Profiles: Describes the shape or form of a product as the quality characteristic. Involves measurements in two or three dimensions.

Phase I: Linear Profile Establishment

Goal: Use historical data to establish a stable and representative baseline for the linear profile. This involves identifying and removing assignable causes from past data.

Parameter Estimation

- For each historical profile, parameters (β0 and β1) are estimated using Ordinary Least Squares regression

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- Estimators are needed because true parameters are unknown.
- OLS estimators are unbiased and have minimum variance, making them Best Linear Unbiased Estimators

Variance-Covariance Matrix

- This matrix (denoted as S) is crucial as it defines the expected "wobble" and correlation of the profile's parameters when the process is stable.

$$\begin{pmatrix} S_0^2 & S_{01}^2 \\ S_{01}^2 & S_1^2 \end{pmatrix}$$

- This allows control charts to distinguish between normal variation and actual shifts

Retrospective Analysis

- Control charts (like a modified Hotelling's T^2 chart) are applied to historical parameter estimates.
- This iterative process ensures the final baseline is based on truly in-control data
- Purpose is to identify and remove outlying historical profiles that could bias the baseline.

Phase II: Online Linear Profile Monitoring

Goal: The primary goal of this phase is the real-time detection of process disturbances as quickly as possible

Control Statistic Calculation and Plotting

- New profiles are collected sequentially as the process operates;
- For each new profile, its parameters are estimated, and a **control statistic is calculated** based on its deviation from the established Phase I baseline
- This statistic is then **plotted on the control chart**.
- **Signal generation and interpretation** is when the plotted point is compared against these control limits.

Advanced Approaches in Phase II Monitoring

- Challenges:
  - Correlation (within/between profiles): Traditional methods often fail due to independence assumption.
  - Model misspecification: Parametric models can bias estimates; non-parametric models may inflate variance.
- Advanced Approaches:
  - U statistic methods address between-profile autocorrelation.
  - Semi-parametric approaches (MRR2) combine parametric and non-parametric fits to overcome model limitations.
  - Robust methods (like Multivariate CUSUM) show superior detection and accuracy across varying model misspecification levels.

Control Charts : Hotelling's T^2 Chart

- Hotelling's T^2 chart is preferred as it simultaneously monitors multiple correlated parameters. It's widely used in multivariate quality control , and considered a good, simple option for linear profile monitoring.

$$T^2 = (\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_0)^T \hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}_0}^{-1} (\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_0)$$

- $\hat{\boldsymbol{\beta}}_i$ Vector of estimated parameters (intercept, slope) for the current (new) profile i.
- $\hat{\boldsymbol{\beta}}_0$ Known in-control mean vector of parameters (target intercept, slope) from Phase I.
- $\hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}_0}^{-1}$ Inverse of the estimated in-control variance-covariance matrix from Phase I. It weights deviations and accounts for parameter interdependencies.

- Distribution: T^2 eventually behaves like an approximate chi-squared distribution χ_p^2 . Where p is the number of parameters
- Upper control Limit of this chart is calculated using this formula: $UCL = \chi_{\alpha,p}^2$

Control Charts : Alternative Charting Strategies

- **Univariate EWMA Charts (after X-coding)**
  - Kim, Mahmoud, and Woodall (2003) propose coding independent variables (X-values) so their average becomes zero. In simpler terms, by shifting the X-axis so its center is at zero, This transformation leads to a new form of the linear model:

$$Y_{ij} = B_0 + B_1 X_i' + e_{ij}$$

  - $B_0 = \beta_0 + \beta_1 \bar{X}$ is the new intercept
  - $X_i' = X_i - \bar{X}$ are the centered independent variable values.
  - Three separate EWMA charts created to monitor Y-intercept, slope, and error variance, offering easier diagnosis of process changes.
- **Residual-Based Control Charts**
  - This approach involves analyzing the residuals (differences between observed and predicted values from the fitted profile model). It detects process shifts through changes like increasing variability in residuals.
  - Control charts can monitor summary statistics of residuals like Root Mean Squared Error.
  - Residuals can be further utilized in non-parametric or semi-parametric approaches for fitting and monitoring.

Performance Measures

- The Average Run Length (ARL) is a key performance measure for control charts. It quantifies, on average, how many samples (or profiles) are expected to be collected before a control chart signals.

- By analyzing ARL values, particularly comparing in-control average run length and out of control average run length for different shift sizes and types, researchers and practitioners can effectively compare the performance of various control chart designs and optimize their parameters for specific monitoring goals

In-Control ARL

- Represents the average number of samples expected when the process is in control before a false alarm occurs.
- A higher ARL is desirable to indicate a lower false alarm rate. Studies often aim for an ARL0 of 200.

Out-of-Control ARL

- Represents the average number of samples collected after a true shift has occurred before the chart signals.
- A lower ARL is highly desirable for quicker detection of real process changes.

Practical Application: R Simulation Dataset Setup

- For this practical illustration, a synthetic dataset representing linear profiles was generated.
- The underlying true linear model for each observation yi at independent variable xi is given by: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

```
phase1 _ data $y <- true _ beta0 _in_ control + true _ beta1 _in_ control *
phase1 _ data $x +  rnorm ( m1_ phase1 * n , mean = 0 , sd = true _ sigma _ error )
```

- ***Global Parameters (R Code Definitions):***

  - **`n = 10:`** *Number of observations (x-points) per profile.*
  - **`x_values = 1:n :`** *Fixed independent variable values for all profiles.*
  - **`true_beta0_in_control = 5`** *: True in-control intercept* β_0
  - **`true_beta1_in_control = 2 :`** *True in-control slope* β_1
  - **`true_sigma_error = 0.5`** *: Standard deviation of the error term*
  - **`alpha = 0.0027`** *: The Type I error rate (probability of a false alarm). This value is chosen as it directly determines the UCL for the control chart, aiming for a high in-control ARL*
  - **`shift _ start _ profile <- 11`** *: This is the point where the shift starts in Phase II*
  - **`shifted _ beta1 <- 2.5`** *: This is the new shifted value*

Practical Application: R Simulation Phases I Setup

- **Parameter Estimation:** For each of the m1 = 30 historical profiles, a linear regression model (lm(y x, data = df)) was fitted. The lm() function in R, by default, utilizes the Ordinary Least Squares method for parameter estimation.

```
# fit linear model to each phase I profile
 phase1 _ models <- lapply ( split ( phase1 _data , phase1 _ data $ profile _id ) ,
                              function (df) lm( y ~ x , data = df) )
# extract parameter estimates ( beta0 _hat , beta1 _hat) for each phase I profile
beta _ hats _ phase1 <- t( sapply ( phase1 _ models , coef ) ) # transpose to get param est x param matrix
```

- **Variance-Covariance Matrix:** From this collection of m1 individual parameter estimates, the overall variance-covariance matrix of estimators was calculated

```
Sigma _ beta _0_hat <- cov( beta _ hats _ phase1 )
```

- **Retrospective Analysis:** A Hotelling's T^2 chart was applied to the Phase I data to perform retrospective analysis. This involves plotting the T 2 statistic for each historical profile against a calculated UCL. The UCL for Phase I is then determined using the qchisq function based on the chosen α (0.0027) and the number of parameters p (2).

```
# p is the number of parameters being monitored
 p <- 2
UCL_T2 <- qchisq (1 - alpha , df = p )
```

Practical Application: R Simulation Phases II Setup

- **Introducing the shift:**

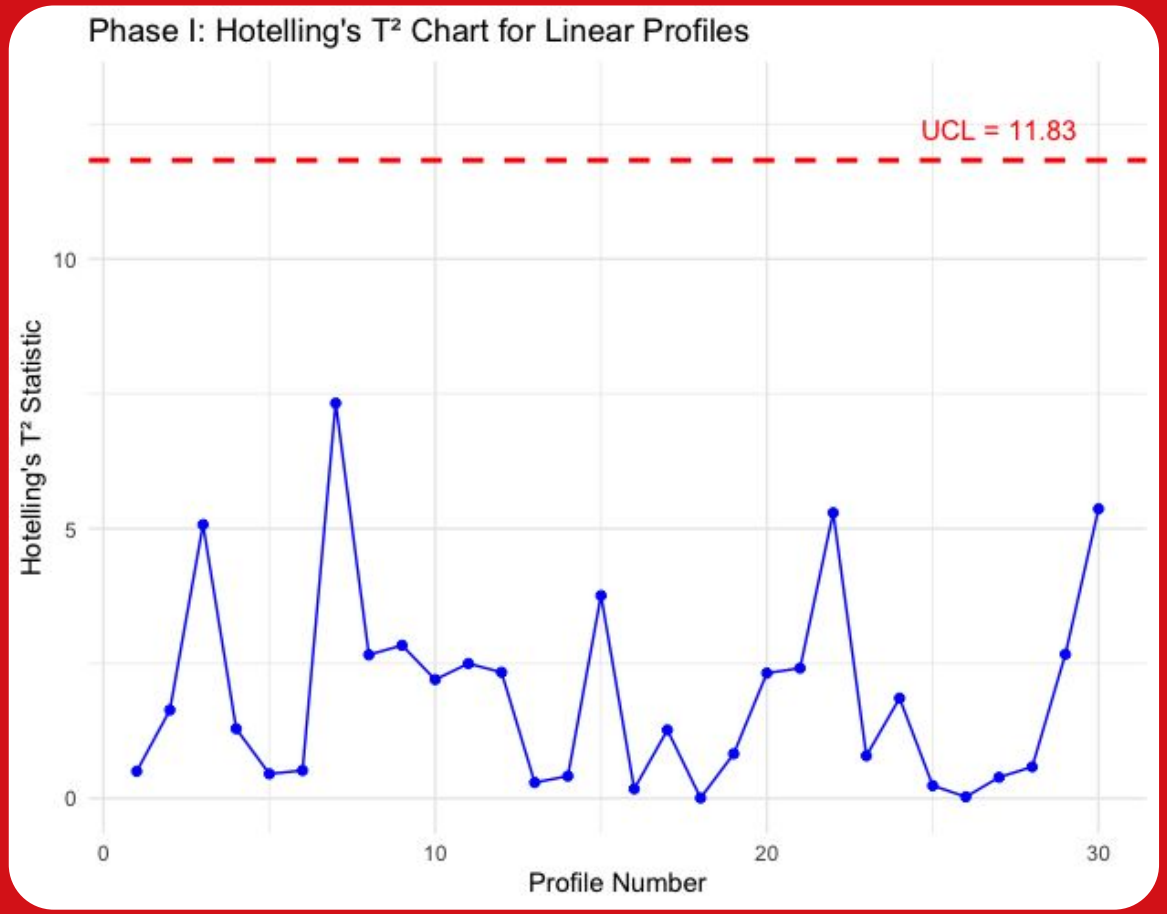
```
for ( i in 1: m2_ phase2 ) {
      current _ beta0 <- true _ beta0 _in_ control
      current _ beta1 <- ifelse ( i >= shift _ start _ profile , shifted _beta1 ,
                          true _ beta1 _in_ control )
 start _row <- ( i - 1) * n + 1
 end _row <- i * n
 phase2 _ data $y [ start _row:end_row] <- current _ beta0 + current _ beta1 *
      phase2 _ data $x [ start _row:end_row] +
 rnorm (n , mean = 0 , sd = true _ sigma _ error ) }
```

- **Estimation for New Profiles**: For each new profile, its parameters are estimated using the same linear model process as Phase I

- **Control Statistic Calculation**: For each new profile, a control statistic is calculated. This calculation uses the new profile's estimated parameters, compared against the in-control parameters (`beta _0_ hat_ overall`) and their variance-covariance matrix (`Sigma _ beta _0_ hat`) established in Phase I.

```
T2_ phase2 _ values <- numeric ( m2_ phase2 )
 for ( i in 1: m2_ phase2 ) {
      beta _ diff <- matrix ( beta _ hats _ phase2 [i , ] - beta _0_ hat_ overall , ncol = 1)
       T2_ phase2 _ values [ i ] <- t( beta _ diff ) %*% solve ( Sigma _ beta _0_ hat) %*% beta _ diff }
```

R Simulation Results: Phase I



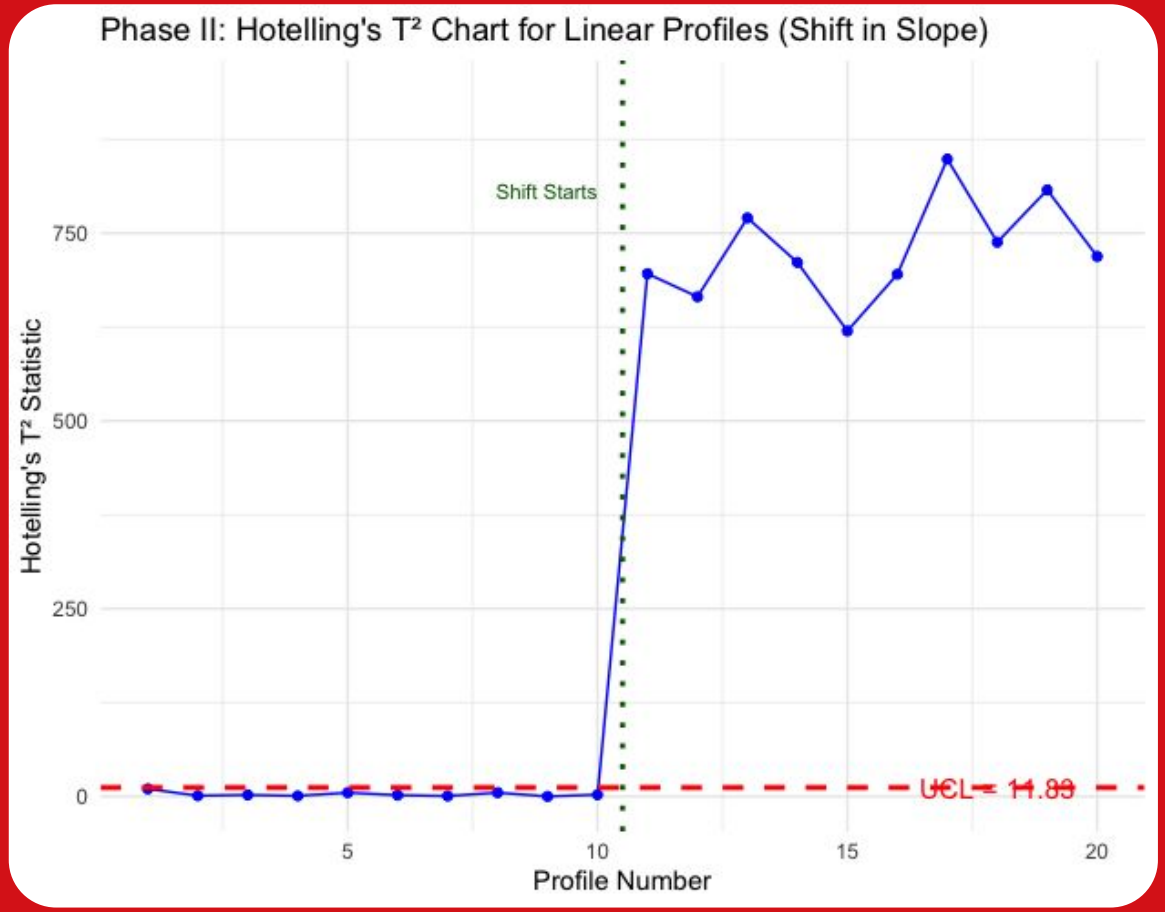
This plot displays the Hotelling's T^2 statistic for each of the 30 simulated Phase I profiles.

The blue points represent the calculated values for each profile. The red dashed line indicates the UCL at 11.83.

All points on this chart fall below the UCL, confirming that the historical data used to establish the baseline was indeed in statistical control.

This retrospective analysis validates the selected in-control state of the process.

R Simulation Results: Phase II



Pre-Shift Behavior (Profiles 1-10):
For the first 10 profiles, which were generated under in-control conditions, the Hotelling's statistic values remain very low, fluctuating close to zero and well below the UCL.
This indicates that the process was stable and in control before the shift.

Post-Shift Behavior (Profiles 11-20):
Immediately after the shift in the slope was introduced at profile 11, the T^2 statistic values exhibit a dramatic increase. The point at Profile Number 11 itself shows a value significantly above the UCL, generating a clear out-of-control signal. The code generates:

```
[Out-of-control signals detected at profiles: 11 12 13 14 15 16 17 18 19 20 ]
```

R Code

```r
#libraries
library(ggplot2)

#global Parameters
set.seed(123)
n <- 10        # number of observations per profile
x_values <- 1:n # fixed observation values for all profiles

# true in-control parameters
true_beta0_in_control <- 5
true_beta1_in_control <- 2
true_sigma_error <- 0.5 # standard deviation of the error term

#phase I data generation and establishment
m1_phase1 <- 30

# creating  a data frame to store phase I data
phase1_data <- data.frame(
  profile_id = rep(1:m1_phase1, each = n),
  x = rep(x_values, m1_phase1)
)
# generating the y values for phase I profiles
phase1_data$y <- true_beta0_in_control + true_beta1_in_control * phase1_data$x +
  rnorm(m1_phase1 * n, mean = 0, sd = true_sigma_error)
```

R Code

```r
# phase I parameter estimation and control Limit calculation
# fit linear model to each phase I profile
phase1_models <- lapply(split(phase1_data, phase1_data$profile_id),
                        function(df) lm(y ~ x, data = df))

# extract parameter estimates (beta0_hat, beta1_hat) for each phase I profile
beta_hats_phase1 <- t(sapply(phase1_models, coef)) # transpose to get m1 x p matrix

# calculatingg the overall in-control parameter vector (mean of Phase I estimates)
# this is our estimate for beta_0 in the T^2 formula
beta_0_hat_overall <- colMeans(beta_hats_phase1)

#calculating the estimated in-control covariance matrix of the parameter estimates
# this is the estimate for Sigma_beta_0 in the T^2 formula
Sigma_beta_0_hat <- cov(beta_hats_phase1)

#calculate Hotelling's T^2 for for retrospective analysis
T2_phase1_values <- numeric(m1_phase1)
for (i in 1:m1_phase1) {
  # (beta_hat_i - beta_0_hat_overall)
  beta_diff <- matrix(beta_hats_phase1[i, ] - beta_0_hat_overall, ncol = 1)

  #T^2 = (beta_diff)^T %*% solve(Sigma_beta_0_hat) %*% beta_diff
  T2_phase1_values[i] <- t(beta_diff) %*% solve(Sigma_beta_0_hat) %*% beta_diff
}
```

R Code

```r
# determine Hotelling's T^2 control limit for Phase II
# p is the number of parameters being monitored
p <- 2
alpha <- 0.0027 # standard alpha

UCL_T2 <- qchisq(1 - alpha, df = p)

# phase I plot Hotelling's T^2 for the retrospective analysis
plot_df_phase1 <- data.frame(Profile = 1:m1_phase1, T2 = T2_phase1_values)
ggplot(plot_df_phase1, aes(x = Profile, y = T2)) +
  geom_line(color = "blue") +
  geom_point(color = "blue") +
  geom_hline(yintercept = UCL_T2, linetype = "dashed", color = "red", linewidth = 1) +
  annotate("text", x = max(plot_df_phase1$Profile) * 0.9, y = UCL_T2 * 1.05,
           label = paste("UCL =", round(UCL_T2, 2)), color = "red", size = 4) +
  labs(title = "Phase I: Hotelling's T² Chart for Linear Profiles",
       y = "Hotelling's T² Statistic", x = "Profile Number") +
  theme_minimal() +
  ylim(0, max(T2_phase1_values, UCL_T2) * 1.1)

# phase II data generation with a shift
m2_phase2 <- 20 # number of profiles in phase II
shift_start_profile <- 11 # shift starts at this profile number
shifted_beta1 <- 2.5    # new shifted value
```

R Code

```r
# create a data frame for phase II
phase2_data <- data.frame(
  profile_id = rep(1:m2_phase2, each = n),
  x = rep(x_values, m2_phase2)
)

# generate y values for phase II profiles, introducing a shift
for (i in 1:m2_phase2) {
  current_beta0 <- true_beta0_in_control
  current_beta1 <- ifelse(i >= shift_start_profile, shifted_beta1, true_beta1_in_control)

  start_row <- (i - 1) * n + 1
  end_row <- i * n

  phase2_data$y[start_row:end_row] <- current_beta0 + current_beta1 *
    phase2_data$x[start_row:end_row] + rnorm(n, mean = 0, sd = true_sigma_error)
}

# phase II: monitoring and plotting
# fit linear model to each Phase II profile
phase2_models <- lapply(split(phase2_data, phase2_data$profile_id),
                        function(df) lm(y ~ x, data = df))

# extract parameter estimates for each ehase II profile
beta_hats_phase2 <- t(sapply(phase2_models, coef))
```

R Code

```r
# calculate Hotelling's T^2 for phase II profiles
T2_phase2_values <- numeric(m2_phase2)
for (i in 1:m2_phase2) {
  beta_diff <- matrix(beta_hats_phase2[i, ] - beta_0_hat_overall, ncol = 1)
  T2_phase2_values[i] <- t(beta_diff) %*% solve(Sigma_beta_0_hat) %*% beta_diff}

# phase II: plot Hotelling's T^2
plot_df_phase2 <- data.frame(Profile = 1:m2_phase2, T2 = T2_phase2_values)
ggplot(plot_df_phase2, aes(x = Profile, y = T2)) +
  geom_line(color = "blue") +
  geom_point(color = "blue") +
  geom_hline(yintercept = UCL_T2, linetype = "dashed", color = "red", linewidth = 1) +
  geom_vline(xintercept = shift_start_profile - 0.5, linetype = "dotted", color = "darkgreen",
  annotate("text", x = max(plot_df_phase2$Profile) * 0.9, y = UCL_T2 * 1.05,
           label = paste("UCL =", round(UCL_T2, 2)), color = "red", size = 4) +
  annotate("text", x = shift_start_profile - 1, y = max(T2_phase2_values, UCL_T2) * 0.95,
           label = "Shift Starts", color = "darkgreen", hjust = 1, size = 3) +
  labs(title = "Phase II: Hotelling's T² Chart for Linear Profiles (Shift in Slope)",
       y = "Hotelling's T² Statistic", x = "Profile Number") +
  theme_minimal() +
  ylim(0, max(T2_phase2_values, UCL_T2) * 1.1)

cat("Phase II Monitoring Results")
out_of_control_profiles <- which(T2_phase2_values > UCL_T2)
if (length(out_of_control_profiles) > 0) {
  cat("Out-of-control signals detected at profiles:", out_of_control_profiles)
} else { cat("No out-of-control signals detected in Phase II.")}
```

Conclusion

- Profile monitoring is crucial for managing quality in modern processes where product characteristics are defined by functional relationships rather than single measurements.
- This paper has provided a theoretical overview of linear profile monitoring, detailing its two distinct phases: Phase I, dedicated to establishing a stable in-control baseline, and Phase II, focused on real-time detection of process disturbances
- The practical application through R simulations demonstrated the effectiveness of the Hotelling's T2 control chart in successfully identifying shifts in linear profile parameters.

> *"Profile monitoring is a tool of the future to match the technology of today that will realize more widespread acceptance as it becomes more readily accessible"*

Noorossana, Saghaei, & Amiri (2011)

Thank you